

Introduction and Notation

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May 2026

1 Introduction

This series of lectures and lecture notes aims to be a somewhat comprehensive review of mathematical ideas relevant to aerospace engineering. The student can use this material either to familiarize themselves with these mathematical concepts, or as an introduction if the concepts are unfamiliar. All information here is self contained within these lectures and notes, with reference to wider mathematical ideas. Suggestions are made throughout of where more information can be found on different topics.

One recurring source is Wikipedia. Wikipedia is very good for looking up definitions of mathematical ideas and theorem statements. That said, Wikipedia can be very technical in notation, as will these notes to some extent. There is a comprehensive coverage of mathematical notation at the end of this introduction, and any notation questions should be answered there. The reason for including this notation is that it is never taught in engineering and is widely used in STEM fields. Notation can often be a barrier to understanding different papers or books written from other disciplines. I offer the opportunity here for aerospace engineering students to become comfortable with this semi-standard notation.

The format of this collection of concepts is as follows:

The videos stand alone and should be used as the primary source of information. The notes cover the same topics as the videos, but where the videos just cover example problems the notes are cover the topics more comprehensively.

The general structure of all other lectures and notes in this amalgamation of mathematical information is first, an introduction into the topic and reminder of the connections to other topics relevant to understanding of the topic at hand. Then comes formal definitions, and theorem statements. Following the definitions will be examples, and worked out problems concerning the given topic. Then to conclude, connections to some engineering applications will be covered.

Note that these lectures and notes present mathematics in the form familiar to what is used in engineering. However, there are generalizations and formalizations of a lot of these definitions and theorems, that generalize these ideas to broader mathematical structure. For example these lectures and notes will do calculus over the real numbers $\mathbb{R}$ not

over the complex numbers $\mathbb{C}$. That is not to say these ideas are not valid for the complex plane, but instead that the way in which the definitions and theorems are presented requires adjustment in mathematical technicality to be valid over the complex plane. With that said engineering is done over the complex plane, not just the real numbers. It is just not worth formalizing these ideas in that case due to the depth at which engineering uses the complex plane versus the technicality required to truly do these generalizations justice.

I do still encourage anyone who is interested to pursue theses or any other topics discussed in more depth. I point you first to Wikipedia as that's usually a good starting place for math, then looking for online textbooks in addition to YouTube videos that discuss these topics. Do be aware that YouTube or other online videos are generally not super formalized but can give the high level concepts. Depth you will find in text books. And just google blank topic text books. They are there!

Why would we concern ourselves with formal mathematical definitions? The benefit to learning the formal definition is not only a better understating of the concepts, but also an outline of the possibility and limitations of this concept. For example by understating the definition of continuous and differentiable we can conclude why piecewise functions are not differentiable at a point of discontinuity.

To iterate both the notes and lectures will cover the same topics in two distinct formats and provide more places to find different perspective will be suggested in the notes.

2 Introduction to Set Theory

This is included, as set theory forms much of the foundation of to what mathematics engineers use. So to better understand notation, and concepts along with definitions we will need a little set theory. With that said, to perform any of the operations or to learn through example, set theory is not necessary; however if we are to understand what the math is doing we need set theory. The reason for seeking an understanding of what the math is doing is to allow broader applications of what we know. If we know what something is, we can better draw relations between that and other ideas.

An excellent example of this comes from using differential equations to model changing concentrations of volumes using systems of differential equations.

A set is loosely speaking a collection of objects. An example could be a marker a pen and a pencil, call this set A , or another example call it set B could be the integers written as $\mathbb{Z}$. Set $B = \mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\} = \{x \in \mathbb{R} | x \text{ is an integer}\}$ is the collection of all integers. Sets can be finite like set A or infinite like set B .

A subset is a set contained within another set. For example the set of a pen and marker is a subset of the set A . The union of two or more sets is the set of all the elements in the sets in the union. The intersection of two or more sets is the set of all elements in common across the collection of sets.

3 Notation

Firstly note that not all notation is consistent, and these notes and lectures will use mathematical notation. But there is conflicting notation between physics and engineering. These differences will also be outlined here.

Second, this is not a comprehensive list of all symbols. Rather this list is comprehensive to and beyond the scope of these lectures and notes and enough to read other disciplines. But what this will lack is both basic notation such as a definition of an equal sign, addition symbol and thing of that nature as well as more advanced concepts that are highly specialized.

Relations:

Symbol	Name	Meaning
$\forall$	for all	for all x in Y
$\exists$	there exists	there exists x in Y
$\subset$	subset	X is a subset of Y
$\subseteq$	subset	X is a subset of Y and X may or may not equal Y
$\in$	an element of	x is an element of the set Y
$\cap$	intersection	the set of all elements in both X and Y
$\cup$	union	the set of all elements in X and all elements in Y
$\wedge$	and symbol	$A \wedge B$ is read A and B
$\vee$	or symbol	$A \vee B$ is read A or B
$\mathbb{R}$	real numbers	the set of real numbers
$\mathbb{C}$	the set of complex numbers	$\mathbb{C} = \{a + bi a, b \in \mathbb{R}\}$
$\mathbb{N}$	natural numbers	the set of non negative integers, sometimes including or not including 0
$\mathbb{Z}$	the set of integers	$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$
$\times$	cartesian product	$A \times B = \{(a, b) a \in A, b \in B\}$
A^n	short hand for cartesian product	$A \times A \times \dots \times A$, n times
$\implies$	implies	statement P implies statement Q
$\iff$	if and only if	statements P and Q imply each other
$:=$	defined to be	defined to be in mathematics and sometimes physics
$\equiv$	defined to be	defined to be in engineering and sometimes physics
$\cong$	congruent (in math and sometimes physics)	the same as

Additionally, any thing with a slash through it means not that. For example $\neq$ means not equal. Or $\nexists$ means there does not exist x in Y .

4 Common Definitions

Well Defined

Well defined means that a definition for a mathematical object is unambiguous and makes logical sense in the context in which the object is defined.

Sets

A set is a gathering of distinct well defined elements, brought together under a single whole.

Lists

A List is a gathering of well defined ordered elements.

Subsets

A subset $S \subseteq U$ is a set such that all the elements $x \in S$ are also elements $x \in U$.

Note that if $S \subseteq U \implies$ if $x \in S$ then $x \in U$ but the other direction is not necessarily true so $y \in U$ can be $y \notin S$.

Union

The union of two or more sets is the set of all elements in the collection of sets. So if $A_0, \dots, A_n$ are sets then the union is

$$\cup_{i=0}^n A_i = A_0 \cup A_1 \cup \dots \cup A_n = \{x | x \in A_0 \vee x \in A_1 \vee \dots \vee x \in A_n\}$$

where if $x \in A_i$ for any $0 \leq i \leq n$ then $x \in A_0 \cup A_1 \cup \dots \cup A_n$.

Note if $x \in A_0$ and $x \in A_n$ then $x \in A_0 \cup A_1 \cup \dots \cup A_n$ but x does not appear in $A_0 \cup A_1 \cup \dots \cup A_n$ more than once. Similarly if $x \in A_0$ and $x_n \notin A_n$ then we still have $x \in A_0 \cup A_1 \cup \dots \cup A_n$.

Intersection

The intersection of two or more sets is the set of elements in common in the collection of sets in the intersection.

So if $A_0, \dots, A_n$ are sets then the intersection is

$$\cap_{i=0}^n A_i = A_0 \cap A_1 \cap \dots \cap A_n = \{x | x \in A_0 \wedge x \in A_1 \wedge \dots \wedge x \in A_n\}$$

meaning $x \in A_0 \cap A_1 \cap \dots \cap A_n \iff x \in A_i, \forall 0 \leq i \leq n$

So in short, the union has every element in all the sets in the union. And the intersection is the set of all the elements that are in every set.

In looking at the cartesian product we will examine the case where we have two sets in the product then the case where we have n sets in the product. The first case of two sets is a simplification of the case with n sets. I recommend using the second case, and using the first case as a teaching example.

Cartesian Product (of two sets)

The cartesian product is the set of all ordered pairs of elements in two sets. Let A and B be sets. Then define the cartesian product $A \times B$ as

$$A \times B = \{(a, b) | a \in A \wedge b \in B\}$$

This is the set of points (a, b) where a is an element of A and b is an element of B .

Note that given that the cartesian product requires order pairs. this means the the elements of the cartesian product are lists not sets.

Cartesian Product (of n sets)

The cartesian product is the set of all ordered pairs of elements in two sets.

Mappings or Function (of n sets)

Let A and B be sets. Then we define a mapping (or function) $f : A \rightarrow B$ as the process of assigning the elements of A to elements in B .

Note functions and mappings share the same definition in mathematics. This is brought up when looking at other resources in STEM fields where you may come across a resource using the word mapping instead of function.

Explicit Equations

Consider a relation between two variables x, y . Say y is dependent on x and x is independent. Then $y = f(x)$.

We define an equation of this form to be an explicit equation.

Implicit Equations

Consider a relation between two variables x, y . If the relation can not be solved or is not solved for x in terms of y or visa versa we call this an implicit equations and is of the form $f(x, y) = 0$

Even Odd Functions

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ and $x \in \mathbb{R}$. Then

f is said to be an even function if $f(-x) = f(x)$

f is said to be an odd function if $f(-x) = -f(x)$